

Image Realism Enhancement Method Based on Improved Neural Neighbor Style Transfer

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Abstract—Object detection technology is a very important technique that relies on a large amount of data for training. However, obtaining sufficient data for specific detection targets can be challenging. In cases where data are limited, it is necessary to increase the number of datasets using various methods. We propose an improved algorithm based on the Neural Neighbor Style Transfer (NNST) algorithm to better adapt to the task of realistic style transfer. Through comparative experiments, we find that image realism could be enhanced when the alpha value is 0.75 and content loss, color correction, and high pixel output are used. Additionally, we design a more effective feature extraction network called SE-VGG19. Compared to the original VGG16, SE-VGG19 can improve the network's ability to perceive style and extract features, making the generated images match the target style better while preserving the original content features. Furthermore, we suggest using the center cosine distance instead of the original Euclidean distance for loss measurement. After comparison and verification, our method has been proven to improve image realism compared to the original algorithm greatly.

Keywords—Image Style Transfer, Image Realism Enhancement, Neural Neighbor Style Transfer, SE-VGG19

I. INTRODUCTION

Object detection technology constantly evolves with the further development of science and technology. When training object detection models, we need a large number of data for training and validation to ensure the accuracy of their training results. However, the particularity of some images leads to the scarcity of such images, and the available data are limited when training models. In order to increase the number of data, some people have used Unreal Engine 4 (UE4) software to model the target objects that need to be detected and integrate them into various scenes to increase the image data. The dataset generated by UE4 modeling can indeed alleviate the problem of limited training data to some extent, but there is still a certain gap compared to images in real scenes. Therefore, enhancing the image realism of the dataset obtained from virtual modeling to enable better training

of the object detection model is an urgent problem that needs to be solved.

Style transfer can perform style transformation on the virtual scene images generated from UE4 modeling. Consider synthetic and real images as two styles, with the synthetic domain as the source domain and the real domain as the target domain. By learning the style of the real domain and transforming the images in the synthetic domain, the realism of the synthesized images can be improved, making them closer to real data. This will alleviate the problem of poor training effect of the object detection network due to the large difference between the virtual dataset and the real dataset to a certain extent, so that the object detection network can show better detection ability on the real image data after training. When faced with difficulties in manually collecting datasets leading to insufficient training data for object detection networks and poor training results for virtual synthetic data, using style transfer to perform style transformation on virtual synthetic data to enhance its realism can make the data closer to real scenes, thereby better improving the training effectiveness of object detection networks. At the same time, this will also enable the object detection network to better adapt to various real-world scenarios, enabling it to perform well even in unseen real-world scenarios, improving its effectiveness and reliability in practical applications.

In 2015, Gatys et al. innovatively proposed a style transfer algorithm based on convolutional neural networks [1]. By fitting the features of image data and using a pre-trained VGG model as the image feature extractor, they explicitly distinguished the abstract features of image content and style and successfully transformed an image into another style, achieving significant results. In 2016, Gatys et al. continued to extend existing neural transformation styles by introducing control over spatial position, color information, and cross-spatial scales, combining multiple different style information to generate new attractive styles from existing styles [2]. However, the algorithm based on neural network style transfer proposed by Gatys et al. introduces the distortion phenomenon. In 2017, Luan et al. proposed the first deep learning-based

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realistic style transfer algorithm based on Gatys [4,20]. In order to enhance the realism of image reconstruction, two strategies are introduced. One is to add an image realism regularization parameter to constrain the use of local affine color changes in the content image in the reconstructed image to prevent distortion. Another approach is to perform semantic segmentation on content images, which can improve the realism of the resulting images. Their algorithm has been successfully applied in various application scenarios such as delay transformation, weather, season, and artistic style. However, Luan et al.'s algorithm often produces inconsistent stylization, and this algorithm is expensive. In 2018, Li et al. proposed a new and faster realistic image stylization method to address the issue of spatially inconsistent stylization and significant artifacts in several methods for stylizing realistic images [5]. They proposed using pooling and anti-pooling layers with coordinate information to replace the pooling and upsampling layers in WCT, thereby improving the problem of spatial information loss. In the same year, Park et al. introduced a new style attention network (SAnet) that can effectively and flexibly integrate local style patterns based on the semantic spatial distribution of content images while maintaining content structure without losing the richness of style [7]. In 2022, Kolkin et al. proposed Neural Neighbor Style Transfer (NNST) based on Neural Style Transfer (NST). They designed two variants, NNST-D and NNST Opt, achieving faster speed and higher output quality, respectively [9].

In our research, we primarily focus on enhancing synthetic images' realism by improving neural style transfer algorithms. Through comparative experiments, we discover that incorporating content loss, color correction, and high pixel output can significantly enhance the realism of the output image. Then, we introduce a new network architecture called SE-VGG19, which replaces the original VGG16 for feature extraction. This change enhances the network's ability to extract style features during style transmission while better preserving the content features of the original image. Additionally, we refine the loss function by suggesting the use of center cosine distance instead of the original Euclidean distance to measure feature similarity. Based on these improvements, we compared the effectiveness of our improved algorithm with the original NNST algorithm in enhancing image realism, demonstrating that our improvement is effective.

II. METHODS

Neural Neighbor Style Transfer (NNST) is an improved algorithm based on neural style transfer, which better improves the visual quality of the output image. However, NNST is commonly used to transfer artistic styles. In order to achieve image realism enhancement and improve the effect of realistic style transfer based on NNST, we have proposed improved NNST.

A. Implementation Principle

Style transfer based on NNST mainly includes four parts: feature extraction, feature matching, image optimization, and

color post-processing. In feature extraction truncation, it utilizes a pre-trained feature extraction network VGG16. Feature extractor $\Phi(x)$ extracts features by extracting the hypercolumns formed by activating the convolutional layers in the first four blocks of the pre-trained VGG16 when transmitted to x . In VGG, each output layer can be used to represent features. The higher the level, the more abstract the extracted features are, while the lower the level, the more specific the extracted features are. NNST extracts features from style and content images and zero centers content and style features. When the size of content image C is $H_c \times W_c$, and the size of style image S is $H_s \times W_s$, the stylized output is represented as:

$$T \in R^{\frac{H_c}{4} \times \frac{W_c}{4} \times 2688} \quad (1)$$

Assuming that $\Phi'(x)$ is a function extracted from the input image x and its rotation, where a single feature vector is represented as $\Phi'(x)_j$, then the feature vector is represented as:

$$T_i = \underset{\Phi'(S)_j}{\operatorname{argmin}} D(\Phi(C)_i - \mu_C, \Phi'(S)_j - \mu_{S'}) \quad (2)$$

In equation (2), D is the cosine distance, μ_C is the average feature extracted from the content image, and $\mu_{S'}$ is the mean feature extracted from the style image and its rotated copies.

In the feature-matching stage, NNST uses two feature-matching mechanisms. Firstly, it uses the hypercolumn matching mechanism for initialization. When it reaches the final stage, the feature splitting matching method is adopted to ensure good content preservation and improve the high-frequency matching between the output and style images. The features of different layers are mixed and matched from different positions in the style image. When synthesizing images, using a learned decoder G , decoder G is responsible for taking $T \in R^{\frac{H_c}{4} \times \frac{W_c}{4} \times 2688}$ as input and feeding T to four independent branches, each responsible for generating a level of four Laplacian pyramids. Afterward, the output image is synthesized by considering the outputs of the four branches as the level of the Laplacian pyramid and combining them appropriately. In the image optimization stage, NNST finds the output image x by using the Adam optimizer to minimize the following objectives:

$$\min_x - \frac{1}{P} \sum_0^{P-1} \cos(\Phi(x)_i, T_i) \quad (3)$$

In equation (3), T is the given target feature, Φ is the feature extractor, $P = W_c H_c / 16$ is the number of spatial locations in T and $\Phi(x)$.

Finally, NNST will perform color post-processing on the image. When converting an image to LAB mode, the brightness generated by NNST-D or NNST-Opt is used, but the AB channels of the image are discarded and replaced with the AB channels of the original content. In order to match the AB channel of the input content image with the L channel of the generated output image, this algorithm performs bilateral filtering on the AB channel guided by the L channel. It matches the mean and covariance of the color distribution formed by

the output L channel and the filtered AB channel with the color distribution of the style image, thereby obtaining an output image that better preserves the texture color distribution characteristics.

B. Feature Extraction Network

In our proposed improved NNST, we have designed a more effective feature extraction network called SE-VGG19. Compared to VGG16, VGG19 has a deeper architecture and can better capture advanced features of images. Compared to VGG16, it adds additional convolutional layers, which not only allows it to use more layers of feature representations to match and adjust styles during style transfer, thereby better preserving the content information of the original image but also helps capture broader contextual information due to its relatively large receptive field size, thus better understanding the texture and structure in the image. In style transfer tasks, it can better extract and express style information. The Squeeze and Excitation (SE) module is an attention module that enhances convolutional neural networks [18]. It enhances the expressive power of channels in the network by adaptively weighting them. It consists of two main steps, namely Squeeze and Excitation. The compression process compresses the spatial information of the input feature map through global average pooling, and the excitation process combines the learned channel attention information with the input feature map to ultimately obtain a feature map with channel attention. We adopt VGG19, which has a deeper network depth and introduced the SE attention module in VGG19. As shown in Fig. 1, we add an SE attention module after the convolutional layer to further enhance the network's ability to extract style features based on VGG19. At the same time, by automatically learning the weights of each channel, the SE module can make the network pay more attention to maintaining the content information of the original image when stylizing the image, avoiding excessive distortion of the content.

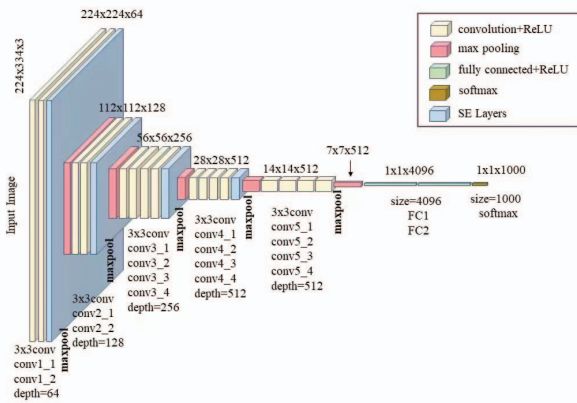


Fig. 1. Network structure diagram of SE-VGG19.

C. Loss Function

In addition, distance functions are needed to measure feature similarity in style transfer. In the original algorithm, when

not using content loss, the distance used is cosine distance. When using content loss, the L2 distance will be used to pre-calculate the content self-similarity matrix while measuring feature similarity using cosine distance. In the following experimental section, it will be mentioned that using content loss can help enhance image realism. However, L2 distance refers to the straight-line distance between two points, which focuses more on the absolute difference in data. The cosine distance focuses more on the relative difference between two vectors in direction. Equation (4) reflects the definition of cosine similarity.

$$\begin{aligned} \text{similarity} &= \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} \\ &= \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \end{aligned} \quad (4)$$

Among them, A_i and B_i are the components of vectors A and B , respectively. The center cosine distance is an improvement on the cosine distance, which first centers the vector and then calculates the cosine distance of the centered vector. The center cosine distance can reduce the translation effect between vectors through centralized processing and pay more attention to the angle differences between vectors. Therefore, we propose using the center cosine distance instead of the original Euclidean distance to pre-calculate the content self-similarity matrix when using content loss in NNST style transfer.

III. EXPERIMENT

A. Network Parameters

To achieve the transfer of image realism style, the selected content images in this experiment are virtual scene images generated by UE4 modeling, and the style images are real-world images of vehicles driving on the road. All experiments in this paper were carried out on the NVIDIA DGX-1 server, equipped with the GPU model of Tesla V100 and 12G video memory. During the debugging process of this experiment, it was found that when $lr = 1e^{-3}$, using the bilinear function as the Laplacian pyramid interpolation function resulted in a better network generation effect.

In addition, we studied the alpha value that determines the degree of stylization, whether to use content loss, color correction, and the selection of max_scls values. The entire experimental process was conducted by controlling variables. As shown in Fig. 2, the alpha value determining the degree of stylization is not the higher, the better. When the alpha value is set to 0.75, a balance between stylization and content retention is achieved.

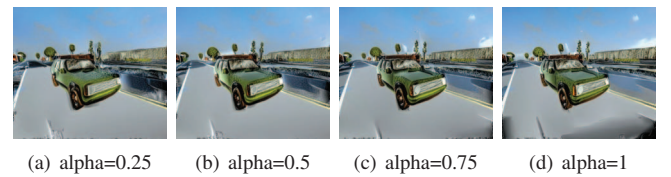


Fig. 2. The generation effects of different alpha values.

Fig. 3 compares image generation results when using different processing methods. Through experiments, we found that using content loss has an improved effect on the transfer of realistic style, resulting in better preservation of the content structure of the generated image and a more realistic color scheme. When color correction is not used, the color area will experience significant deviation. It can be seen that color correction is an essential operation in the realistic style transfer based on NNST. In order to further improve the output image pixels, this experiment sets the number of stylized scales max_scls from 4 to 5. The final generation result shows that when max_scls is set to 5, although some artifacts are generated, the details of the image are better preserved, the overall image is clearer, the details are better drawn, the pixels are improved, and the generated image is more realistic.

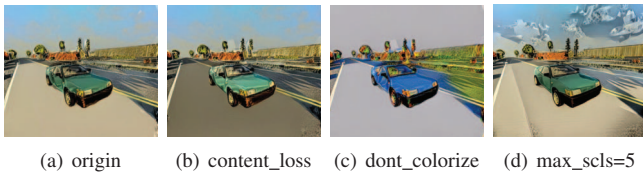


Fig. 3. The generation effects of different uses.

Based on the above experiments, we combine the parameters and usage that are conducive to realistic style transfer, namely setting $\alpha = 0.75$, learning rate $lr = 1e^{-3}$, using bilinear interpolation, using content loss and color correction, and obtaining a comparison with the original results. It is found that the improved and optimized NNST style transfer algorithm produces significantly better results than the original results, greatly reducing distortion and artifacts in the image. Its generated images are more natural in color, light, and shadow and more realistic. On this basis, we added the method of using high pixel output to generate images, and the comparison results are shown in Fig. 4. It can be seen that the improved generated effect is significantly better than the original result image, and the results using high pixel output are even better. The details in its image are more precise, and the overall image has higher pixels and a more realistic feel. Comparing the result of high pixel output with the result before the improvement, we find that the realism of the image using high pixel output is improved.

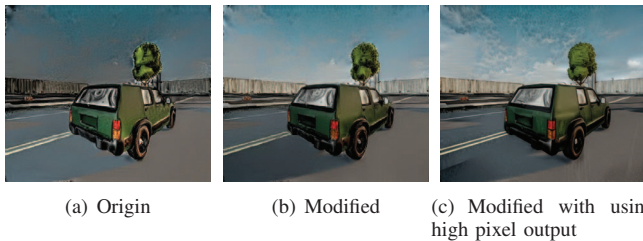


Fig. 4. Comparison of two optimization methods and the original results.

B. Feature Extraction Networks SE-VGG19

As mentioned in methods, we have replaced the feature extraction network. We replaced the original VGG16 network with our proposed SE-VGG19 network for feature extraction. The results of using the original network parameters and loss function for style transfer are shown in Fig. 5.



Fig. 5. Comparison of results using VGG16 (first line) and SE-VGG19 (second line) as feature extraction networks.

Fig. 5 shows that its generation effect significantly improves when using our proposed feature extraction network SE-VGG19 instead of the original VGG16 network. The colors in its generated images are more realistic, especially the shadows of vehicles on the road, which are more in line with the colors of the real world. In addition, the distortion and artifacts in the image have also been improved to a certain extent. Therefore, using our proposed SE-VGG19 for feature extraction can make the colors in the generated image more realistic and better preserve content features while ensuring stylization.

C. Loss Function

This section investigates the impact of using L2 distance and cosine distance on the NNST style transfer algorithm. In the original algorithm, the distance used when not using content loss was the cosine distance. When using content loss, the L2 distance will be used to pre-calculate the content self-similarity matrix under the premise of using the cosine distance. As mentioned before, in order to improve the realism style transfer effect based on NNST, we use content loss in the experiment. Therefore, in this section, L2 distance will be replaced with center L2 distance, standard cosine distance, and center cosine distance for comparison, as shown in Fig. 6.

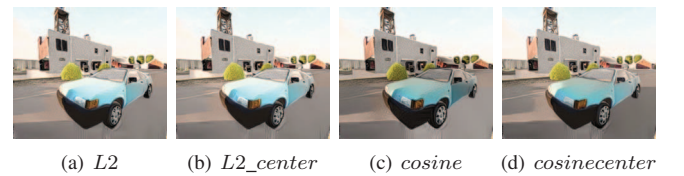


Fig. 6. The generation effects of using different distances.

In Fig. 6, there is little difference between using the L2 distance and the L2_center distance. When cosine distance is used, the overall color of the picture is more realistic. When the center cosine distance is used, the car's shadow in the generated graph is more real and complete, and the content

is better preserved than the result generated by the cosine distance. In order to further verify which distance to pre-calculate the content self-similarity matrix can improve the performance of the style transfer algorithm when using content loss, we compare the loss curves generated by the above four methods on the same graph, as shown in Fig. 7.

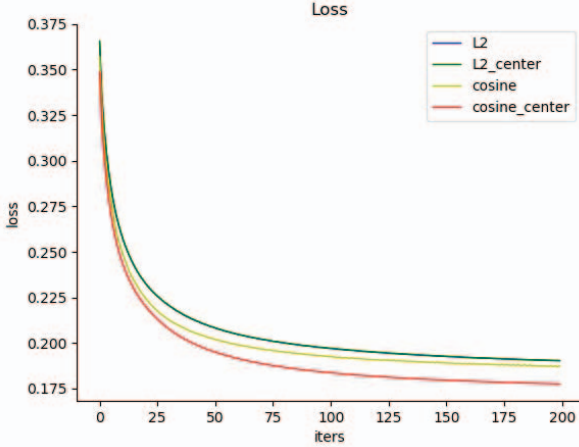


Fig. 7. Comparison of loss using different distances.

From Fig. 7, it can be seen that the loss curve converges with an increase in the number of iterations. The loss curves of the L2 and L2_center basically overlap, and their loss values are significantly higher than those generated when using cosine distance. The loss value generated by the center cosine distance is significantly lower than the standard cosine distance. It can be seen that using center cosine distance to calculate the content self-similarity matrix can be more conducive to image style transfer, making the style difference between the generated image and the style image smaller.

D. Validation of Improvement

Based on the experimental results mentioned above, we combine methods conducive to enhancing image realism. These methods include using good hyperparameter settings, content loss, color correction, high pixel output, and the SE-VGG19 feature extraction network we proposed. Additionally, we employed the center cosine distance to calculate content loss. The generated results are shown in Fig. 8. As can be seen, our improvements have further enhanced the image generation effect, preserving more content details in the image and enhancing the overall color realism of the road surface shadows and the image. Compared to the original generated image, it is clear that our improved algorithm can better enhance the realism of the image in style transfer tasks.

To further validate the effectiveness of the improvements made in this paper, we used the Learned Perceptual Image Patch Similarity (LPIPS) metric for evaluation [19]. LPIPS, also known as perceptual loss, is used to measure the perceptual difference between two images. It extracts image features by using pre-trained deep neural networks (such as



Fig. 8. Comparison between the original algorithm (first line) and our improved one (second line).

VGG and AlexNet) and calculates the differences between these features to evaluate the similarity between images. Its core idea is to learn the reverse mapping from generated images to real images, thereby forcing the generator to learn to reconstruct the features of real images from generated images and prioritize the perceptual similarity between them. LPIPS is more in line with human perception than traditional methods such as L2/PSNR, SSIM, and FSIM. The lower the value of LPIPS, the more similar the two images are, and vice versa, the greater the difference. Our results are shown in Table I.

TABLE I
COMPARISON OF LPIPS VALUES BEFORE AND AFTER IMPROVEMENT

	image1	image2	image3	image4	image5
Content	0.6531	0.6806	0.6945	0.7464	0.6865
Origin	0.6250	0.6450	0.6549	0.7406	0.6757
Our improved	0.6086	0.6268	0.6270	0.7120	0.6441

Table 1 shows that the LPIPS value is significantly reduced after our improvement, indicating that the improved NNST algorithm we proposed can better enhance image realism in the style transfer task.

IV. CONCLUSION

This paper achieves the goal of enhancing image realism by improving the neural neighbor style transfer. We first conduct a comparative experiment and find that image realism could be better enhanced when the alpha value is 0.75 and content loss, color correction, and high pixel output are used. Afterward, we propose a more effective feature extraction network, SE-VGG19. Compared to the original VGG16 feature extraction network, we deepen the network depth and introduce the SE attention module, enabling it to enhance the network's ability to extract style features while better preserving content information. In addition, we propose to use the center cosine distance instead of the original Euclidean distance to pre-calculate the content self-similarity matrix when using the content loss. Finally, we compare the proposed improvements with the original generated results, proving that our proposed improvements can better enhance image realism in style transfer tasks.

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